Closed Loop Control of Single-phase Boost PFC Converter

Introduction

In the realm of power electronics, the design and implementation of power factor correction (PFC) converters are paramount for efficient energy utilization. Following the critical phase of component sizing and selection, which ensures the physical realization of a single-phase boost PFC converter with appropriate ratings for switches, inductors, capacitors, and diodes, the next crucial step is to establish robust control over its operation. This lecture delves into the intricacies of closed-loop control, transforming a well-designed hardware system into an intelligent, responsive power converter.

The journey from theoretical component selection to a functional converter necessitates a control strategy that can dynamically manage its behavior. Without a sophisticated control mechanism, the converter would merely operate in an open-loop fashion, unable to adapt to varying input conditions or load demands. This lecture focuses on how closed-loop control enables the converter to achieve its primary objectives: maintaining a stable output voltage and drawing a unity power factor current from the AC source, thereby ensuring optimal performance and compliance with power quality standards.

Core Concepts

Term	Definition	Significance
`Duty Ratio (d)`	The fraction of the switching period during which a switch (e.g., S1) is in the ON state.	It is the primary control variable, directly influencing the average voltage applied to the inductor and thus the current and output voltage.
`Switching Frequency (fsw)`	The rate at which the converter's switches (S1, S2) are turned ON and OFF.	Kept constant and much higher than line frequency to simplify control, reduce EMI, and ensure stable operation.
`Control Objectives`	The desired operational goals for the converter,	These define what the closed-loop system must achieve to

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	specifically output voltage regulation and unity power factor current draw.	meet performance specifications.
`Average Current Control`	A control strategy that regulates the average inductor current by adjusting the duty ratio of the switches.	A common and effective method for PFC converters, ensuring the input current follows the input voltage waveform.
`Voltage Controller`	A feedback controller (e.g., PI, P) that processes the error between the reference and feedback output voltage.	Generates a signal that dictates the magnitude of the inductor current required to maintain the desired output voltage.
`Current Controller`	A feedback controller that processes the error between the reference and feedback inductor current.	Adjusts the duty ratio to ensure the actual inductor current closely tracks its reference, achieving unity power factor.
`Saturation Block`	A functional block that limits the output of the voltage controller to prevent exceeding component current ratings.	Essential for protecting the converter's components from overcurrents and ensuring safe operation within design limits.
`Small-Signal Modeling`	A technique used to derive transfer functions by linearizing the converter's non-linear equations around an operating point.	Crucial for designing stable and responsive controllers by providing a mathematical representation of the plant's dynamic behavior.
`Transfer Function (Gi(s))`	Represents the dynamic relationship between a change in duty ratio and the resulting change in inductor current.	Essential for designing the current controller, as it describes how the inductor current responds to control inputs.

Term	Definition	Significance
`Transfer Function (Gv(s))`	Represents the dynamic relationship between a change in inductor current and the resulting change in output capacitor voltage.	Essential for designing the voltage controller, as it describes how the output voltage responds to changes in inductor current.

Detailed Analysis

1. Control Objectives and Variables

The fundamental purpose of closed-loop control in a single-phase boost PFC converter revolves around two critical objectives:
* **Output Voltage Regulation**: The converter must maintain a constant DC output voltage (V naught) at a predefined desired level, irrespective of input voltage fluctuations or load variations. This desired value must inherently be greater than the peak input voltage (Vs,peak). For instance, if the specification demands a 400V output, the control system must ensure this voltage is consistently achieved.
* **Unity Power Factor Current Draw**: Simultaneously, the converter must draw a current from the AC input source that is in phase with the input voltage, thereby achieving a unity power factor. This is crucial for minimizing reactive power and improving overall system efficiency.

To achieve these objectives, the control system must manipulate specific variables within the converter. While passive components like inductors (L) and capacitors (C) are fixed, and diodes (D1-D4) conduct based on their forward bias, the only available control variables are the switching times of the active switches, S1 and S2. Specifically, the **duty ratio (d)**, which is the fraction of the switching period (Ts) during which a switch is ON, becomes the primary means of control. By adjusting the duty ratio, the average voltages applied across the inductor can be controlled, which in turn regulates the inductor current and ultimately the output voltage. A key assumption for simplified control is that the switching frequency (fsw) is kept constant and significantly higher than the line frequency, preventing issues like EMI and complex harmonic interactions.

2. The Average Current Control Scheme

A commonly employed control strategy for PFC converters is average current control. This method aims to regulate the average inductor current by appropriately

changing the duty ratio of the switches. The overall closed-loop control architecture can be visualized as follows:

- **Voltage Control Loop**:
- A **V naught reference** (desired output voltage) is compared with the **V naught feedback** (sensed actual output voltage).
- The error signal is fed into a **Voltage Controller** (e.g., PI controller). This controller's output indicates the required magnitude of the inductor current to correct the voltage error. For example, if the feedback voltage is lower than the reference, the controller will demand a higher inductor current to charge the capacitor and raise the voltage.
- The output of the voltage controller then passes through a **Saturation Block**. This block is crucial because it limits the maximum current demand to prevent exceeding the physical current ratings of the converter's components (switches, inductor, diodes), which were determined during the sizing phase.
- The saturated output, representing the peak inductor current magnitude ($I_{L,pk}$), is then multiplied by the **modulus of the unit sine function** (modulus of $\sin(\omega t)$). This multiplication ensures that the reference inductor current waveform tracks the shape of the input AC voltage, which is essential for achieving unity power factor. The modulus of $\sin(\omega t)$ can be generated using a single-phase PLL or by sensing and processing the grid voltage. The result is the **IL reference** (average inductor current reference).
- **Current Control Loop**:
- The **IL reference** is compared with the **IL feedback** (sensed actual inductor current).
- The error signal is fed into a **Current Controller**. This controller's role is to ensure that the actual inductor current closely follows the reference current.
- The output of the current controller, often scaled by a gain (k_i), directly determines the **duty ratio (d)**.
- This duty ratio is then sent to a **Modulator Block**, which generates the Pulse Width Modulation (PWM) signals for switches S1 and S2. Typically, the PWM for S2 is the logical NOT of the PWM for S1. These PWM signals drive the gate drivers of the respective switches.

The entire control logic, from voltage and current controllers to the modulator, is typically implemented within a microcontroller in the digital domain. Sensors are used to provide the V naught feedback and IL feedback to the respective control loops.

3. Transfer Functions and Small-Signal Modeling

To effectively design and tune the voltage and current controllers, it is imperative to understand the dynamic behavior of the converter itself. This understanding is captured through **transfer functions**, which mathematically describe how the converter responds to changes in its control inputs. Since power converters are inherently non-linear, a technique called **small-signal modeling** is employed to derive these transfer functions.

The process of small-signal modeling involves:
1. **Deriving the Average Large-Signal Model**: This involves formulating the state equations of the converter (e.g., for inductor current and capacitor voltage) in the time domain, representing their average behavior over a switching cycle.
2. **Applying Small Perturbations**: Small AC variations (perturbations) are introduced to the steady-state operating point of the system's inputs, such as the duty ratio (d) and input voltages.
3. **Linearization**: The non-linear state equations are linearized around the steady-state operating point. This simplifies the analysis by allowing the use of linear control theory.
4. **Conversion to S-Domain**: The linearized time-domain equations are then transformed into the S-domain (Laplace domain) to obtain the transfer functions.

Two key transfer functions are derived:
* **Gi(s) = iL(s) / d(s)**: This transfer function describes how the inductor current (iL) responds to a change in the duty ratio (d). It captures the dynamics of the inductor current loop.
* **Gv(s) = V naught(s) / iL(s)**: This transfer function describes how the output capacitor voltage (V naught) responds to a change in the inductor current (iL). This is because changes in inductor current affect the converter current (i_conv), which in turn affects the current flowing into the capacitor (i_c) and thus the capacitor voltage.

Knowing Gi(s) is crucial for designing the current controller, while Gv(s) is essential for designing the voltage controller. These transfer functions allow engineers to select appropriate controller parameters (e.g., proportional and integral gains for a PI controller) to ensure stability, desired response time, and accurate tracking of reference signals.

Key Takeaways

- [ ] Closed-loop control is essential for regulating output voltage and achieving unity power factor in PFC converters.

- [ ] The duty ratio of the switches (S1, S2) is the primary control variable in a boost PFC converter.
- [ ] The switching frequency is kept constant and high to simplify control and minimize EMI.
- [ ] Average current control is a common strategy, involving both an outer voltage loop and an inner current loop.
- [ ] The voltage control loop generates a reference for the inductor current magnitude based on output voltage error.
- [ ] A saturation block is critical to limit the current reference, protecting components from overcurrents.
- [ ] The inductor current reference is shaped by multiplying its magnitude with the modulus of the input sine wave to achieve unity power factor.
- [ ] The current control loop ensures the actual inductor current tracks its reference by adjusting the duty ratio.
- [ ] Small-signal modeling is used to derive transfer functions (Gi(s) and Gv(s)) that characterize the converter's dynamic behavior.
- [ ] These transfer functions are vital for designing and tuning the parameters of the voltage and current controllers for optimal performance.

Conclusion

The implementation of closed-loop control is the cornerstone of modern power converter design, transforming a static hardware assembly into a dynamic, intelligent system capable of precise power management. By meticulously regulating the output voltage and ensuring a unity power factor, the single-phase boost PFC converter not only delivers stable power but also contributes significantly to grid efficiency and power quality. The average current control scheme, coupled with the analytical rigor of small-signal modeling, provides a robust framework for achieving these objectives. Understanding these principles is fundamental for any engineer working with power electronics, as it bridges the gap between theoretical component selection and the practical realization of high-performance, reliable power conversion systems. The insights gained from this lecture lay the groundwork for further exploration into advanced control techniques and their application in diverse charging infrastructure scenarios.